

SURV4CR: Survival Analysis for Clinical Research with R

Full Program, September/October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Survival Analysis for Clinical Research with R

Format: Online, cohort-based

Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours

Dates: 05/10/2026 – 18/10/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)

Time: Morocco time

Price: 150€ (1500 MAD)

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

SURV4CR provides a practical and methodologically rigorous introduction to survival analysis for clinical research using R. It covers the full time-to-event analysis workflow: survival data structure, censoring, Kaplan-Meier estimation, log-rank tests, Cox regression, proportional hazards diagnostics, competing risks, RMST, survival machine learning, dynamic RMST analysis, and temporal phenotyping.

The course emphasizes clinical interpretation, correct handling of censoring, model assumptions, reproducible analysis, and responsible use of machine learning for time-to-event outcomes.

1.2 Target Audience

Clinicians, clinical researchers, epidemiologists, PhD students, and analysts working with follow-up data. Basic R and regression concepts recommended.

1.3 Prerequisites

Basic familiarity with R and regression (as covered in BIOSTATR) is recommended but not mandatory.

1.4 Learning Objectives

By the end of SURV4CR, participants will be able to:

1. define time origin, event, follow-up time, and censoring mechanism;
2. structure survival data using R;
3. estimate and interpret Kaplan-Meier curves and perform log-rank tests;
4. fit and assess Cox proportional hazards models, including diagnostics and extensions;
5. analyze competing risks using cumulative incidence and Fine-Gray models;
6. use RMST as a clinically interpretable survival estimand;
7. fit, evaluate, and interpret survival machine learning models;
8. compute dynamic RMST curves and apply temporal clustering for survival phenotyping;
9. produce a reproducible survival analysis report.

2 Detailed Schedule

Table 1: SURV4CR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	05/10/2026 (Mon)	18h00- 20h00	Survival Data Structure, Study Design & Kaplan-Meier/Log-Rank	Time-to-event outcomes, censoring, Surv() objects, Kaplan-Meier estimator, median survival, log-rank test
2	1	07/10/2026 (Wed)	18h00- 20h00	Cox Proportional Hazards Model	Cox model, hazard ratios, partial likelihood, adjusted models, PH assumption, Schoenfeld residuals
3	1	09/10/2026 (Fri)	18h00- 20h00	Cox Model Extensions	Stratified Cox models, time-varying covariates and effects, frailty models, clustered survival data
4	1	11/10/2026 (Sun)	10h00- 12h00	Competing Risks and Flexible Parametric Models	Cause-specific hazards, cumulative incidence, Fine-Gray model, flexible parametric survival models, cure models
5	2	12/10/2026 (Mon)	18h00- 20h00	RMST as a Clinical Estimand	RMST definition, difference and ratio, choice of time horizon, interpretation under non-proportional hazards
6	2	14/10/2026 (Wed)	18h00- 20h00	Survival ML: Foundations, Evaluation & Tuning	Penalized Cox, random survival forests, oblique forests, IPCW metrics, C-index, Brier score, cross-validation
7	2	16/10/2026 (Fri)	18h00- 20h00	Survival ML Interpretation, Dynamic RMST & Temporal Phenotyping	SHAP, local explanations, dynamic RMST curves with tvrmst , temporal clustering with unsurv
8	2	18/10/2026 (Sun)	10h00- 12h00	End-to-End Survival Analysis Project	Complete workflow: KM, Cox, RMST, survival ML, interpretation, dynamic RMST, phenotyping

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required.

Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.